

## CHERKASY, Ukraine

WMO: 334870

Lat: 49.4092N

Lon: 32.0031E

Elev: 351

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 84-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 2             | -2.7       | 3.3 | -7.6                          | 3.7 | -2.0 | -2.0 | 5.0 | 5.0  | 34.1                  | 22.6 | 26.3 | 29.4 | 5.9                   | 300  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.8                   | 86.8            | 67.1 | 83.9 | 65.8 | 81.3 | 64.7 | 70.0                | 80.8 | 68.3 | 79.1 | 66.9 | 77.3 | 8.2                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 99.2 | 74.3 | 64.7 | 92.6 | 72.3 | 63.0 | 87.3 | 71.5 | 34.2          | 81.0 | 32.7 | 79.2 | 31.6 | 77.7 | 77.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 21.9          | 18.7 | 16.5 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 47.3   | 26.0  | 26.6  | 35.4  | 48.7  | 60.4  | 64.6  | 67.9  | 67.2  | 58.2  | 46.9  | 36.2  | 28.6  |  |  |
|                                                                        | DBStd       | 17.51  | 10.65 | 10.74 | 8.88  | 7.33  | 6.61  | 5.51  | 5.58  | 6.03  | 6.78  | 8.27  | 9.56  | 8.36  |  |  |
|                                                                        | HDD50       | 3223   | 743   | 656   | 456   | 112   | 5     | 0     | 0     | 0     | 11    | 158   | 419   | 663   |  |  |
|                                                                        | HDD65       | 6797   | 1208  | 1076  | 918   | 492   | 174   | 73    | 34    | 46    | 220   | 563   | 865   | 1128  |  |  |
|                                                                        | CDD50       | 2252   | 0     | 0     | 3     | 72    | 328   | 439   | 554   | 533   | 259   | 60    | 4     | 0     |  |  |
|                                                                        | CDD65       | 353    | 0     | 0     | 0     | 2     | 33    | 62    | 123   | 114   | 18    | 1     | 0     | 0     |  |  |
|                                                                        | CDH74       | 3470   | 0     | 0     | 0     | 28    | 257   | 509   | 1172  | 1296  | 206   | 2     | 0     | 0     |  |  |
|                                                                        | CDH80       | 909    | 0     | 0     | 0     | 4     | 29    | 104   | 302   | 428   | 42    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 8.1    | 9.5   | 8.5   | 9.5   | 8.6   | 7.7   | 7.4   | 7.2   | 6.6   | 7.4   | 7.8   | 8.1   | 8.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 22.3   | 1.6   | 1.4   | 1.3   | 1.6   | 1.9   | 2.6   | 2.9   | 2.1   | 1.7   | 1.3   | 1.7   | 1.9   |  |  |
|                                                                        | PrecMax     | 31.3   | 6.3   | 3.4   | 2.6   | 3.9   | 4.5   | 5.9   | 5.8   | 4.7   | 7.0   | 4.7   | 3.3   | 3.9   |  |  |
|                                                                        | PrecMin     | 14.1   | 0.3   | 0.0   | 0.0   | 0.4   | 0.2   | 0.0   | 0.5   | 0.2   | 0.2   | 0.0   | 0.2   | 0.3   |  |  |
|                                                                        | PrecStd     | 4.1    | 1.1   | 0.9   | 0.7   | 0.8   | 1.1   | 1.2   | 1.4   | 1.2   | 1.5   | 1.1   | 0.8   | 0.8   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 46.2  | 49.7  | 62.4  | 78.4  | 83.0  | 86.5  | 88.7  | 91.8  | 84.4  | 72.3  | 59.3  | 48.3  |  |  |
|                                                                        |             | MCWB   | 42.1  | 45.2  | 49.3  | 60.9  | 62.7  | 66.9  | 68.4  | 66.5  | 64.5  | 59.4  | 53.0  | 43.8  |  |  |
|                                                                        | 2%          | DB     | 41.0  | 44.9  | 56.9  | 69.7  | 80.1  | 82.8  | 85.9  | 87.9  | 79.6  | 67.7  | 55.2  | 44.8  |  |  |
|                                                                        |             | MCWB   | 38.6  | 42.2  | 47.2  | 54.6  | 61.9  | 65.9  | 67.5  | 66.7  | 62.6  | 58.4  | 51.2  | 41.6  |  |  |
|                                                                        | 5%          | DB     | 38.9  | 41.2  | 51.6  | 65.9  | 76.9  | 79.8  | 83.0  | 84.6  | 75.5  | 63.5  | 51.5  | 41.1  |  |  |
|                                                                        |             | MCWB   | 36.8  | 38.8  | 44.3  | 52.9  | 62.2  | 64.6  | 66.3  | 65.8  | 62.2  | 56.0  | 47.7  | 38.8  |  |  |
|                                                                        | 10%         | DB     | 37.1  | 38.2  | 47.0  | 61.7  | 73.1  | 76.5  | 80.4  | 81.1  | 71.2  | 59.1  | 47.9  | 38.1  |  |  |
|                                                                        |             | MCWB   | 35.2  | 35.8  | 41.7  | 50.6  | 60.7  | 63.9  | 65.9  | 64.2  | 59.7  | 54.2  | 45.2  | 36.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 42.7  | 45.9  | 51.5  | 60.0  | 67.0  | 70.9  | 73.8  | 71.2  | 67.7  | 63.1  | 54.9  | 45.0  |  |  |
|                                                                        |             | MCDB   | 45.5  | 48.2  | 61.0  | 75.2  | 75.3  | 78.1  | 83.3  | 82.1  | 79.7  | 68.3  | 58.6  | 47.6  |  |  |
|                                                                        | 2%          | WB     | 38.9  | 43.0  | 48.0  | 56.8  | 65.1  | 68.4  | 70.7  | 69.0  | 64.9  | 59.5  | 51.7  | 41.9  |  |  |
|                                                                        |             | MCDB   | 40.2  | 45.2  | 55.2  | 66.6  | 74.6  | 78.3  | 80.7  | 81.7  | 75.1  | 64.7  | 54.8  | 44.4  |  |  |
|                                                                        | 5%          | WB     | 37.1  | 39.1  | 45.0  | 54.2  | 63.6  | 66.8  | 68.8  | 67.5  | 62.9  | 57.0  | 48.1  | 39.0  |  |  |
|                                                                        |             | MCDB   | 38.5  | 41.2  | 50.5  | 63.1  | 73.4  | 76.3  | 79.0  | 80.0  | 72.2  | 62.1  | 50.8  | 40.9  |  |  |
|                                                                        | 10%         | WB     | 35.4  | 36.0  | 42.3  | 52.1  | 61.8  | 65.1  | 67.1  | 65.9  | 61.0  | 54.6  | 45.1  | 36.5  |  |  |
|                                                                        |             | MCDB   | 36.7  | 37.7  | 46.7  | 61.0  | 71.0  | 73.7  | 76.8  | 77.6  | 69.6  | 58.8  | 47.6  | 38.0  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 8.8   | 10.9  | 12.9  | 17.1  | 19.1  | 17.2  | 19.8  | 21.7  | 18.6  | 16.5  | 10.0  | 8.0   |  |  |
|                                                                        | 5% DB       | MCDBR  | 8.2   | 12.0  | 21.3  | 24.8  | 21.9  | 22.8  | 23.5  | 27.0  | 25.8  | 23.8  | 15.9  | 11.2  |  |  |
|                                                                        |             | MCWBR  | 6.8   | 9.3   | 12.8  | 12.6  | 8.8   | 9.6   | 8.9   | 9.8   | 11.8  | 14.6  | 11.8  | 9.1   |  |  |
|                                                                        | 5% WB       | MCDBR  | 8.3   | 12.1  | 19.1  | 22.1  | 19.3  | 18.9  | 20.5  | 23.6  | 23.3  | 19.0  | 13.0  | 10.3  |  |  |
|                                                                        |             | MCWBR  | 7.3   | 9.9   | 12.4  | 12.0  | 9.0   | 9.5   | 9.3   | 9.8   | 12.3  | 12.7  | 10.1  | 8.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.319 | 0.352 | 0.384 | 0.446 | 0.414 | 0.407 | 0.436 | 0.436 | 0.404 | 0.369 | 0.350 | 0.321 |  |  |
|                                                                        | taud        |        | 2.194 | 2.151 | 2.197 | 2.096 | 2.275 | 2.315 | 2.243 | 2.229 | 2.281 | 2.353 | 2.304 | 2.260 |  |  |
|                                                                        | Ebn at Noon |        | 222   | 243   | 258   | 253   | 269   | 271   | 260   | 254   | 249   | 238   | 211   | 205   |  |  |
|                                                                        | Edh at Noon |        | 25    | 33    | 38    | 46    | 41    | 39    | 42    | 40    | 35    | 27    | 23    | 21    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 299   | 551   | 944   | 1412  | 1837  | 1959  | 1906  | 1699  | 1146  | 677   | 278   | 206   |  |  |
|                                                                        | RadStd      |        | 56    | 89    | 146   | 133   | 149   | 156   | 115   | 103   | 168   | 99    | 44    | 50    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A   | N/A   |       |
| (47) Regional (12 neighbors) |  | +1.30 | N/A     | N/A    | +1.84 | N/A     | N/A   | -239 | -333        | +212  | +122  |       |

Nomenclature: See separate page